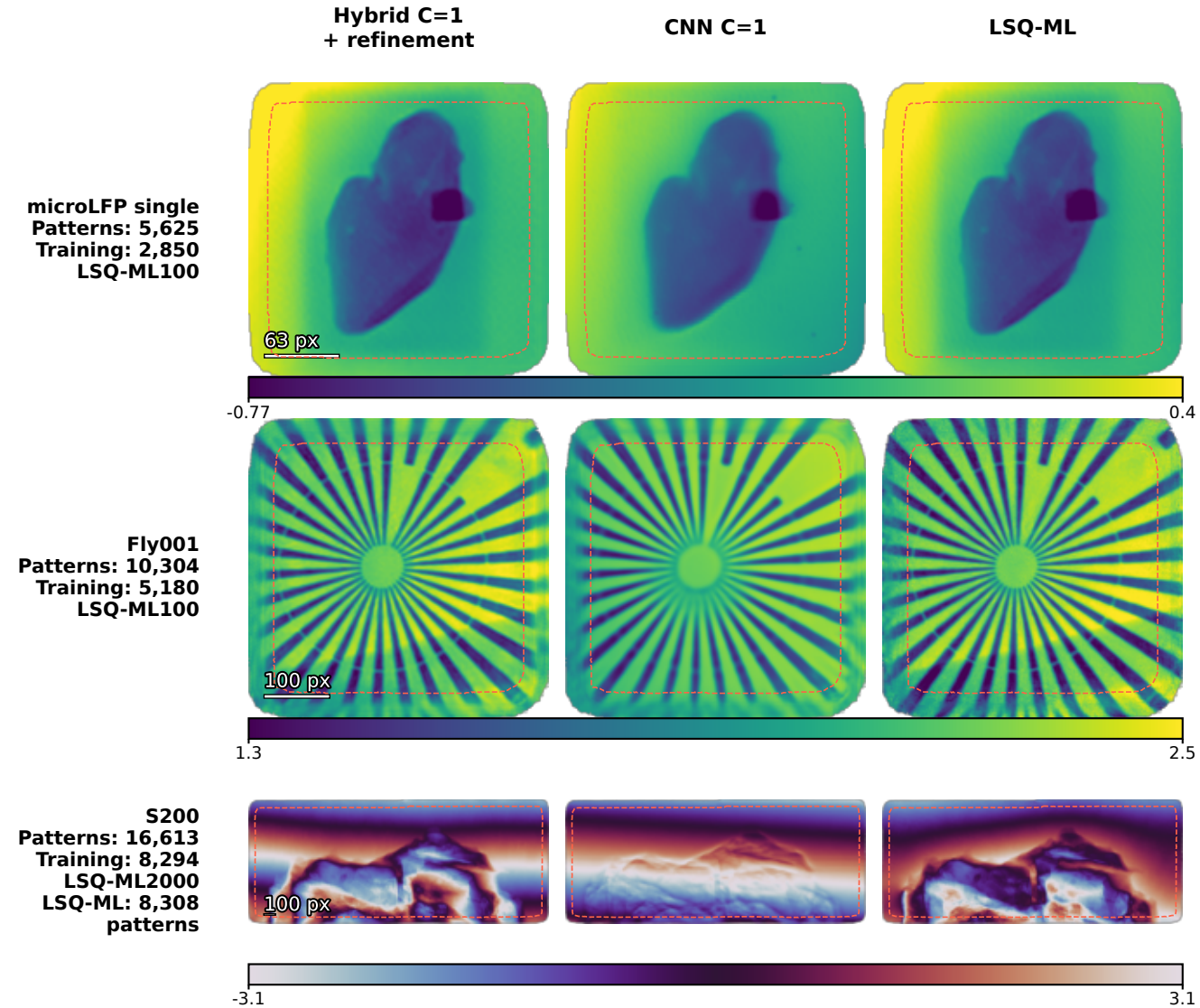


Phase is shown in radians; amplitude shows relative object amplitude. Pattern counts refer to the displayed acquisition and distinct training patterns. High half selects checkpoints (spatial validation, not an independent test; no guard band). Reference is an experimental estimate; dashed contour marks scored interior. Matched measured CNN C=1 is shown without patch-offset alignment, tilt seed, or phase-field refinement. It is an architecture baseline under this study protocol, not a reproduction of the published CNN result. Whole-acquisition images include the training region. Paired groups assign C4 only where it has lower phase MAE after refinement and higher phase SSIM; other datasets use C1. This is the C1-selected group. S200 LSQ-ML2000: historical 8,308-pattern subset versus 16,613 neural patterns; iteration-budget limited, not a matched timing result.